

IT568 – GenAI for Software Engineering

Elective (2-0-4-4), Winter 2026

Instructor: Saurabh Tiwari, 1102, Faculty Block-1

Course Placement: UG/PG

Course format: 2 Lecture hours and 4 Lab hours.

Prerequisite:

- Software Engineering
- Proficiency in at least one programming language (Python/Java/C++).
- Basic understanding of software engineering principles (SDLC, UML, testing).
- Familiarity with machine learning concepts is helpful but not mandatory.

Course Content: This course explores how Generative Artificial Intelligence (GenAI) is transforming the practice of software engineering from requirements analysis and design modeling to coding, testing, documentation, and maintenance. Students will learn to leverage large language models (LLMs) and AI-assisted tools to automate and enhance different phases of the software development lifecycle (SDLC).

By the end of the course, students will design and implement an AI augmented software engineering assistant that applies GenAI principles to real-world programming and engineering tasks.

Textbooks:

1. Ian Sommerville, Software Engineering, 10th Edition, Pearson, 2020.
2. Tom M. Mitchell & Emily M. Bender, Generative AI: Concepts, Tools, and Applications, 2023.
3. Humphrey, W. S. (2021). A Discipline for Software Engineering. Addison-
4. Wesley.
5. OpenAI, Anthropic, Hugging Face – Official Documentation and Developer Guides.
6. Selected IEEE/ACM papers on “LLM-assisted Software Engineering.”

Lab Sessions:

Hands-on sessions will expose students to GenAI-powered software engineering tools such as GitHub Copilot, ChatGPT, Ollama, and, LangGraph, and LangChain. Students will implement AI-assisted code generation, testing, documentation, and UML modeling pipelines.

- Tools Used: Python, Hugging Face Transformers, LangChain, VS Code, GitHub, Streamlit.
- Computing Resources Required for running the course: Google Colab/Lighting.ai

Evaluation Plan:**(Attendance is Mandatory in Lectures and Labs)**

- Assignments & Labs – 15%
- Mid-Semester Exam – 15%
- Lab Examination – 15%
- End-Semester Exam – 20%
- Capstone Project Implementation & Demo – 25%
- Mandatory Attendance in Lecture and Labs – 10%

Course Outcome:

By completing this course, students will:

- Acquire hands-on proficiency in integrating GenAI tools into software workflows.
- Understand theoretical underpinnings of generative systems in SE contexts.
- Contribute to research and innovation in AI-driven software development.
- Ready for emerging roles such as AI Developer, Prompt Engineer, and Software Engineer with AI Skills.

Program Outcomes:

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X	X	X		X	X	X	X	X